

Homework 1

Advanced Applied Math (MAE5003)

Posted: April. 1, 2026

Due Date: April. 14, 2026

Problem 1

Stirling's formula

Recall that in the lecture, we evaluated the integral $I(x) = \int_{-\infty}^{+\infty} e^{-x \cosh t} dt$ ($x \rightarrow \infty$) by defining a new variable τ such that $\cosh t \equiv 1 + \tau^2$, so that we can compute higher order terms $\mathcal{O}(1/x^n)$ for $I(x)$. Now applying the same idea, we can obtain the higher order corrections to the Gamma function $\Gamma(x+1) = \int_0^\infty t^x e^{-t} dt$:

a) Show that $\Gamma(x+1) = x^{x+1} \int_0^\infty e^{-x(s-\ln s)} ds$.

b) Define $s - \ln s \equiv 1 + \tau^2$. Express s as a power series of τ . (hint: use `Series` in `Mathematica` to express τ^2 as a power series of s . Then use `InverseSeries` to invert the power series.) Convert ds to $d\tau$ (as a power series).

c) Substitute in the above $d\tau$ and integrate. Obtain the higher order corrections to Stirling's formula

$$n! \sim n^n e^{-n} \sqrt{2\pi n} \left(1 + \frac{C_1}{n} + \frac{C_2}{n^2} + \dots \right) \quad (1)$$

up to $\mathcal{O}(1/n^2)$.

Problem 2

Projectile problem

In the lecture, we solved the problem of a vertically launched projectile:

$$\begin{aligned} \frac{d^2 x}{dt^2} &= -g \frac{R^2}{(x+R)^2}, \\ x(0) &= 0, x'(0) = v_0, \end{aligned} \quad (2)$$

by rescaling with characteristic time and length scales $t_c \equiv v_0/g$, $x_c \equiv v_0^2/g$, identifying the small parameter $\epsilon \equiv v_0^2/(gR)$ and applying regular perturbation method.

a) Reproduce the steps of the regular perturbation method using `Mathematica` and obtain the result up to $\mathcal{O}(\epsilon)$. Compare the plots of x vs t (or the corresponding dimensionless variables) with different values of ϵ (say, 0.8, 0.001 and 0) and the numerical result (use `NDSolve`).

b) Using the above result, apply algebraic perturbation to obtain the time it takes for the projectile to return to the ground again, to $\mathcal{O}(\epsilon)$. Convert the final result to the dimensional form.

Problem 3

Consider a mass-spring-dashpot system (Fig. 1). The displacement x satisfies

$$m\ddot{x} = F_s + F_d, \quad (3)$$

where m is the mass, F_s is the restoring force of the spring, and F_d is the damping force from the dashpot. The initial condition is $x(0) = 0$ and $x'(0) = v_0$

a) Assuming linear spring $F_s = -kx$ and linear damping $F_d = -c\dot{x}$, with k and c known constants, nondimensionalize Eq. 3. Your answer should contain the parameter $\alpha = c/\sqrt{mk}$.

b) Assume now the spring is nonlinear $F_s = -kx - Kx^3$. This case is known as *Duffing's equation*. Nondimensionalize the problem now. Your answer should contain 2 parameters, α and $\epsilon = Kv_0^2m/k^2$.

c) Explain why the physical meaning of α is the ratio of the period of the undamped oscillator to the damping time scale, and why ϵ is the ratio of the nonlinear to linear component of the spring.

d) For a two-term expansion of the solution of b) of a weakly nonlinear spring (ϵ small), write down the equations and boundary conditions satisfied by those two terms.

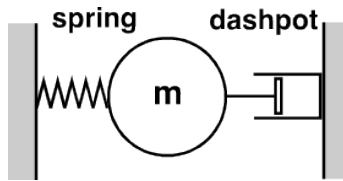


Figure 1: A mass-spring-dashpot system.